

BACKEND STACK

Java 25 · Spring Boot 4.1.1

WEB FRONTEND

Next.js 15 · React 19

CLOUD DATABASE

PostgreSQL 17 (Supabase)

BASE GIT BRANCH

dev (Never main)

1. Prerequisites (Install Before Starting)

Every developer must verify and install the following toolchain on their laptop before cloning:

- **Git:** git-scm.com (Command-line Git).
- **Node.js:** v20+ LTS (nodejs.org) for the Next.js web frontend.
- **Java 25 JDK:** Eclipse Temurin or OpenJDK 25.0.4.1+ (adoptium.net) or via SDKMAN: `sdk install java 25-tem`).
- **IDE:** IntelliJ IDEA (Community or Ultimate) strongly recommended for Spring Boot, or VS Code with the Java Extension Pack.
- **Postman:** postman.com/downloads for automated API testing.

2. Web Frontend Setup (SeekFactory-Web-F)

Repository URL: `https://github.com/Axoraa-tech/SeekFactory-Web-F.git`

Step 2.1: Clone and Switch to 'dev' Branch

```
git clone https://github.com/Axoraa-tech/SeekFactory-Web-F.git
cd SeekFactory-Web-F
git checkout dev
git pull origin dev
```

Step 2.2: Install Dependencies & Configure Environment

Install the required npm packages and create a `.env.local` file in the root folder:

```
npm install
```

Create `.env.local` with the following content:

```
NEXT_PUBLIC_API_URL=http://localhost:8080
```

WHAT THIS DOES:

Directs the Next.js web application to send all REST API calls to your local Spring Boot backend running on port 8080.

Step 2.3: Run the Web Frontend

```
npm run dev
```

Open your browser at: `http://localhost:3000`. The SeekFactory web app will load in development mode.

Step 2.4: Frontend Git Workflow (Branch → Commit → Push → PR)

GOLDEN RULE:

NEVER commit or push directly to 'dev' or 'main'. Always create a new feature branch from 'dev'!

```
# 1. Update dev branch
git checkout dev
git pull origin dev

# 2. Create feature branch
git checkout -b feature/your-feature-name

# 3. Stage and commit changes
git add .
git commit -m "feat: implement product category filter on catalog page"

# 4. Push to remote repository
git push -u origin feature/your-feature-name

# 5. Open Pull Request on GitHub:
# Go to https://github.com/Axoraa-tech/SeekFactory-Web-F
# Click "Compare & pull request"
# CRITICAL: Set BASE branch to 'dev' (base: dev  compare: feature/your-feature-name)
# Submit for team review!
```

3. Backend Setup (SeekFactory-B)

Repository URL: `https://github.com/Axoraa-tech/SeekFactory-B.git`

Step 3.1: Clone and Switch to 'dev' Branch

```
git clone https://github.com/Axoraa-tech/SeekFactory-B.git
cd SeekFactory-B
git checkout dev
git pull origin dev
```

Step 3.2: Critical Rules on Configuration Files

🔥 IMPORTANT CONFIGURATION RULE:

1. `src/main/resources/application.yaml` is provided by default in the repo.
👉 **DO NOT TOUCH OR MODIFY** `application.yaml` ! Keep it as it is.
2. You must **CREATE A NEW FILE** named:
👉 `src/main/resources/application-dev.yaml`

Paste the following into `src/main/resources/application-dev.yaml` :

```
spring:
  datasource:
    url: ${DEV_DB_URL}
    username: ${DEV_DB_USERNAME}
    password: ${DEV_DB_PASSWORD}
    driver-class-name: org.postgresql.Driver
    hikari:
      maximum-pool-size: 3
      minimum-idle: 1
      idle-timeout: 300000
      connection-timeout: 20000
      max-lifetime: 1200000

  jpa:
    show-sql: true

  app:
    cors:
      allowed-origins:
        - http://localhost:3000
        - http://localhost:8081
```

Step 3.3: Configure the Root '.env' File

Create a file named `.env` in the root of `SeekFactory-B` (ensure it is in `.gitignore`):

```
# =====
# SPRING BOOT PROFILE
# =====
SPRING_PROFILES_ACTIVE=dev

# =====
# DEVELOPMENT DATABASE - SUPABASE (Used for local dev)
# =====
DEV_DB_URL=jdbc:postgresql://aws-0-ap-northeast-1.pooler.supabase.com:6543/postgres
DEV_DB_USERNAME=postgres.vvjeqfektsfrfunksslp
DEV_DB_PASSWORD=Axoraa@1967@

# =====
# PRODUCTION DATABASE - SUPABASE (Used ONLY on cloud server)
# =====
PROD_DB_URL=jdbc:postgresql://aws-0-ap-southeast-1.pooler.supabase.com:6543/postgres
PROD_DB_USERNAME=
PROD_DB_PASSWORD=

# =====
# JWT SECURITY KEY & GOOGLE OAUTH
# =====
JWT_SECRET=dGhpc2lzYXZ1cn1zZW1cmVhbmRsb25namd3dHN1Y3JldGtleWZvcnN1ZWtmYWw0b3J5MjAyNgKK
GOOGLE_CLIENT_ID=YOUR_GOOGLE_OAUTH_CLIENT_ID
```

Step 3.4: Run the Backend Locally

- **Terminal (Windows):** `.\mvnw.cmd clean compile` then `.\mvnw.cmd spring-boot:run`
- **Terminal (Mac/Linux):** `chmod +x mvnw && ./mvnw clean compile && ./mvnw spring-boot:run`
- **IntelliJ IDEA:** Set Project SDK to Java 25. Open `AxoraaApplication.java` . Add VM options: `--enable-preview` and Active Profile: `dev` . Click Run.

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```
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=====|_|=====|_/=//_//_  
  
:: Spring Boot ::                (v4.1.1)  
  
INFO ... Starting AxoraaApplication using Java 25.0.4.1 ...  
INFO ... The following 1 profile is active: "dev"  
INFO ... Finished Spring Data repository scanning. Found 14 JPA repository interfaces.  
INFO ... Tomcat initialized with port 8080 (http)  
INFO ... HikariPool-1 - Added connection org.postgresql.jdbc.PgConnection@...  
INFO ... HikariPool-1 - Start completed.  
INFO ... org.flywaydb.core.FlywayExecutor : Database: jdbc:postgresql://aws-0-ap-northeast-1.pooler.supabase.com:6543/postgres  
INFO ... o.f.core.internal.command.DbValidate : Successfully validated 1 migration (execution time 00:00.772s)  
INFO ... o.f.core.internal.command.DbMigrate : Current version of schema "public": 1  
INFO ... o.f.core.internal.command.DbMigrate : Schema "public" is up to date. No migration necessary.  
INFO ... Initialized JPA EntityManagerFactory for persistence unit 'default'  
INFO ... Tomcat started on port 8080 (http) with context path '/'  
INFO ... Started AxoraaApplication in 25.595 seconds
```

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Item	Expected Console Message	Status
1. Active Profile	The following 1 profile is active: "dev"	MUST SEE
2. HikariCP Pool	HikariPool-1 - Added connection ... Start completed.	MUST SEE
3. Supabase DB	Database: jdbc:postgresql://...pooler.supabase.com:6543/postgres (PostgreSQL 17.6)	MUST SEE
4. Flyway Migrations	Successfully validated 1 migration , Schema "public" is up to date.	MUST SEE
5. Tomcat Port	Tomcat started on port 8080 (http) with context path '/'	MUST SEE
6. NO Table Alters	You should NEVER see Hibernate executing ALTER TABLE or CREATE TABLE !	MUST NOT SEE

🚫 **WHY MUST DEVELOPERS NEVER SEE ALTER TABLE IN THE LOGS?**

We use **Flyway Database Migrations** with `hibernate.ddl-auto: validate`. Hibernate is strictly forbidden from modifying cloud database tables directly. If schema changes are needed, write a new migration file (e.g., `V2__add_column.sql`) in `src/main/resources/db/migration/`.

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- `SeekFactory_API_Collection.postman_collection.json`
 - `SeekFactory_Environment.postman_environment.json`
1. Open **Postman** → Click **Import** → Drag and drop both files.
 2. In the top-right environment selector, choose: `SeekFactory - Local / Cloud`.
 3. Run **00. Health & Docs** → **Actuator Health Check** (Expect `200 OK`, `{"status":"UP"}`).
 4. Run **01. Authentication** → **Register Buyer** (Expect `201 Created`). The built-in test script will **automatically capture the JWT access_token into your environment**.
 5. Run **02. User Profile** → **Get User Profile** to test authenticated requests without manual copy-pasting!

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ACTION	WEB FRONTEND (SEEKFACTORY-WEB-F)	BACKEND (SEEKFACTORY-B)
Git Clone	<code>https://github.com/Axoraa-tech/SeekFactory-Web-F.git</code>	<code>https://github.com/Axoraa-tech/SeekFactory-B.git</code>
Base Branch	<code>git checkout dev</code>	<code>git checkout dev</code>
Config Files	Create <code>.env.local</code> (<code>NEXT_PUBLIC_API_URL=http://localhost:8080</code>)	Keep <code>application.yaml</code> untouched; create <code>application-dev.yaml</code> & <code>.env</code>
Active Profile	N/A	<code>dev</code> (loads <code>application-dev.yaml</code>)
Cloud Database	N/A	Supabase PostgreSQL 17 (PgBouncer port <code>6543</code>)
Migrations	N/A	Flyway auto-applies <code>V1__init_schema.sql</code> on startup
Run Command	<code>npm run dev</code> (<code>http://localhost:3000</code>)	<code>.\mvnw.cmd spring-boot:run</code> (<code>http://localhost:8080</code>)
PR Target	Always target <code>dev</code>	Always target <code>dev</code>